



# Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT | SmartInvest – AI-Powered Multi-Asset Investment and Margin Trading Platform |                      |
|----------------------|-----------------------------------------------------------------------------|----------------------|
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### Abstract

SmartInvest is a full-stack, AI-powered multi-asset investment and portfolio management platform designed to provide users with a unified and intelligent environment for managing different types of financial investments. Traditional investment platforms often require users to manage stocks, mutual funds, derivatives, margin requirements, and portfolio analysis through separate interfaces, making investment management complex and time-consuming. SmartInvest addresses this problem by integrating these functionalities into a single platform with an intuitive and modern user interface.

The platform enables users to manage **stocks, mutual funds, and Futures & Options (F&O)** while providing real-time portfolio tracking, transaction management, profit and loss calculations, asset allocation, and performance analytics. It also incorporates a dedicated **margin management and collateral pledging system**, allowing users to monitor available margin, utilized margin, collateral value, and margin requirements. Eligible securities can be pledged as collateral, while appropriate validation mechanisms prevent pledged assets from being incorrectly sold or transferred. These features are designed to simulate important operations found in modern investment and brokerage platforms.

A key component of SmartInvest is its **AI-powered Investment Analyst**, which uses the Google Gemini API to analyze portfolio information and provide personalized, data-driven insights. The AI component can evaluate portfolio diversification, asset concentration, risk exposure, margin utilization, and overall portfolio health. Users can interact with the AI using natural-language queries to better understand their investments and receive educational recommendations based on their portfolio data. This makes the platform more than a conventional trading interface by helping users interpret financial information and make more informed decisions.

The system is developed using a modern **MERN-based technology stack**, with React.js and Vite for the frontend, Tailwind CSS and shaden/ui for the interface, Node.js and Express.js for backend services, and MongoDB with Mongoose for data management. Recharts is used for interactive portfolio analytics and visualization, while JWT and bcrypt provide secure authentication and password protection. REST APIs connect the frontend, backend, database, AI services, and market-data components.

Signature of the Guide

HOD